

DIGIT MEMORY

A study in retrieval as classification.

98.4%

HELD-OUT ACCURACY

1-NN on a stratified 75/25 split, sklearn digits.

STAGE 02A · HASH

3_{us}

EXACT LOOKUP, $O(1)$

bytes-keyed dict.

STAGE 02B · TREE

50_{us}

NEAREST NEIGHBOR, $O(\log n)$

BallTree / KDTree.

STAGE 03 · LABEL

10 classes

DIGITS 0 .. 9

1797 samples.

FIG.01 THE TWO-TIER LOOKUP

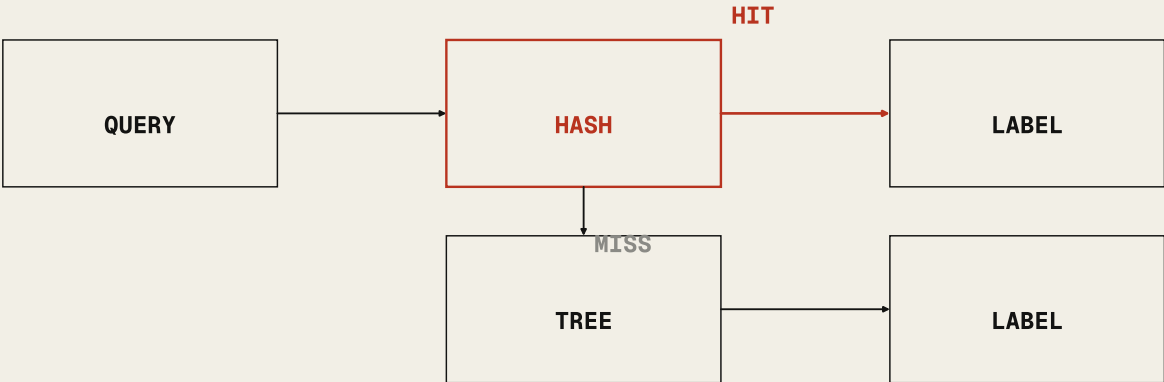
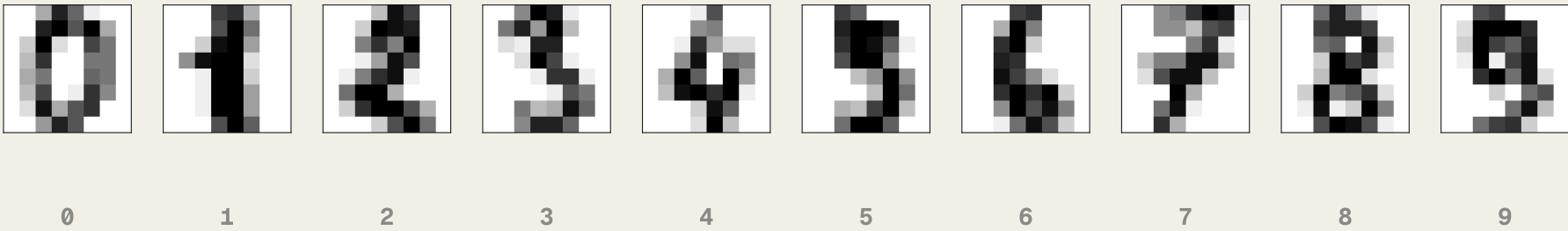


FIG.02 ONE SAMPLE PER CLASS



ROBUSTNESS

Smooth degradation under injected Gaussian noise.

sigma = 0 -> 98.4% sigma = 5 -> 93.8% sigma = 10 -> 67.1%

no cliff, no catastrophic forgetting, no retraining